

1. Demonstrate the validity of $x + yz = (x + y)(x + z)$ by a truth table.
2. Simplify $(a + b + c')(a'b' + c)$ to a minimum number of literals.
3. Simplify $a'bc + abc' + abc + a'bc'$ to a minimum number of literals.
4. Find the complement of $a + a'(b'c + de)$.
5. Draw the logic diagram of $a + a'(b'c + de)$.
6. Implement $F = xy + x'y' + y'z$ using only NAND gates.
7. Obtain the truth table of $(a' + b)(a' + c)$ and express it as sum-of-minterms and product-of-maxterms.
8. Convert $F(x, y, z) = \sum(1, 3, 5)$ to product-of-maxterms.
9. Solve 3.4 (a), (b), (c) and (d) from the textbook.
10. Solve 3.7 (c) from the textbook.
11. Solve 3.9 (a) and (b) from the textbook.

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